



COLLEGE
OF
MEDICINE

Renal & Urinary Tract Problems in the Pregnant woman

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Physiological changes in pregnancy

Renal blood flow & GFR increases by up to 50% by end of 1st trimester.

- As a consequence renal handling of various substances [e.g. glucose] changes.
- Loss of glucose may be x10 non-pregnant by 8-11 weeks – *Lowered Renal Threshold*.
- Similar effect for water soluble vitamins, folic acid, amino acids, electrolytes.



Morphological changes in pregnancy

- Result of hormonal and/or Anatomical [obstructive] factors
- Dilatation of calyces, renal pelvis and ureters starts by 3rd month; reaches maximum by 3rd trimester
- Bladder capacity increases by up to 1000 ml.
- Patient positioning affects renal function with GFR & blood flow being markedly greater in lateral recumbent position rather than supine.



Morphological changes in pregnancy

- These changes will influence the blood and urine concentration of various substances alteration in renal function tests.
- Creatinine clearance increases 110-150 ml/min
- Blood urea decreases ≤ 5 mmol/l [30 mg/dl]
- Mild proteinuria commonly occurs [abnormal if > 300 mg/24 hrs]



Pathological states in pregnancy

May be:

- Present before pregnancy
- Arise during pregnancy
- Arise in postpartum period

Common conditions include:

- **Urinary tract infection**
- Calculus disease -renal colic
- Urinary retention -retroverted uterus
- Acute Renal failure
- Nephrotic syndrome; chronic renal failure



Urinary Tract Infection

A common disorder in pregnancy

Predisposing factors

- Ascending infection from perineal area to bladder.

Affected by:-

- anatomy - short female urethra
- sexual intercourse [Honeymoon cystitis]
- catheterization
- “Obstructive uropathy” of pregnancy
- hormonal changes predisposing to stasis
- pressure of pregnancy unit

UTI in pregnancy: pathogens

- *Escherichia coli* ~75-90% of cases

Virulence assisted by:

- 1.resistance to vaginal acidity,
- 2.rapid division in urine,
- 3.adherence to cells,
- 4.production of chemicals that reduce peristalsis & inhibit phagocytosis



UTI in pregnancy: Clinical picture

1. Asymptomatic

- **Non-significant bacteruria:-** *bacteruria <100,000 bacteria of same species per ml of urine:*
- ~1.5% of these women will develop significant bacteruria during pregnancy
- **Significant asymptomatic bacteruria:-** *bacteruria >100,000 bacteria of same species per ml of urine:*
- ~5% of all pregnant women

UTI in pregnancy: Clinical picture

2. Symptomatic

- Generalised malaise with anaemia without localising features
- anaemia - *normochromic , normocytic resulting from inhibition of erythroblastosis*
- Mild UTI - slightly elevated temperature with urinary disorders possibly including loin tenderness



- **Acute pyelonephritis [*1-2% of women*] - *marked pyrexia* with rigors, loin pain, severe urinary symptoms, very ill patient**



UTI in pregnancy: Obstetric problems

UTI (incl. Asymptomatic bacteruria) associated with:-

- Prematurity
- fetal growth retardation
- increased perinatal mortality & morbidity



Pyelonephritis

- **Pyelonephritis** also occurs in 1% to 2% of pregnant people, most commonly in the second trimester.
- Pyelonephritis is a common cause of serious infections, including septic shock, in pregnant patients. It is the cause of most medical hospitalizations during pregnancy.
- 3.5% of antepartum admissions were due to UTI.
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Risk factors for pyelonephritis in pregnancy

- include obesity, low socioeconomic status, young age, nulliparity, diabetes, smoking, and a history of recurrent UTIs. As with ASB, some patients may be predisposed to infection and may report a history of having had ASB, cystitis, or pyelonephritis in the past. Pyelonephritis is most often right-sided, but it may be bilateral in up to 25% of cases.



Management

- Identify organism & antibiotic sensitivity
- Increase fluid intake +/- infusion
- Antibiotics - preferably bactericidal drugs; use full dose +/- i/v route
- consider augmentin, cephalosporins
- **contraindicated: Septrin, tetracyclin, streptomycin, kanamycin**
- Subsequent follow-up to exclude persisting subclinical infection or predisposing disease (eg. nephrolithiasis; congenital anomaly)



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Urinary tract infection & asymptomatic bacteriuria during pregnancy		
Condition	Clinical features	Management
Asymptomatic bacteriuria	- Positive urine culture (>100,000 colonies/mL) in asymptomatic patient - Screening usually done at 12-16 weeks gestation - Treatment reduces progression to UTI & complications (eg, preterm birth, low birth weight)	- Nitrofurantoin for 5-7 days - Amoxicillin or amoxicillin-clavulanate for 3-7 days - Fosfomycin as single dose - Avoid fluoroquinolones in all trimesters - Avoid trimethoprim/sulfamethoxazole in 1st & 3rd trimesters
Acute cystitis	- Symptomatic patient (eg, dysuria, urgency) with positive urine culture - Considered a complicated UTI	
Acute pyelonephritis	- Flank pain, nausea/vomiting, fever (>38 C [100.4 F]) &/or costovertebral angle tenderness - May or may not have cystitis symptoms	- Hospitalization & intravenous antibiotics (eg, β-lactams, meropenem) - Avoid aminoglycosides unless other agents cannot be used - Change to 10- to 14-day course of oral antibiotics after afebrile for 24 hours



Thank you